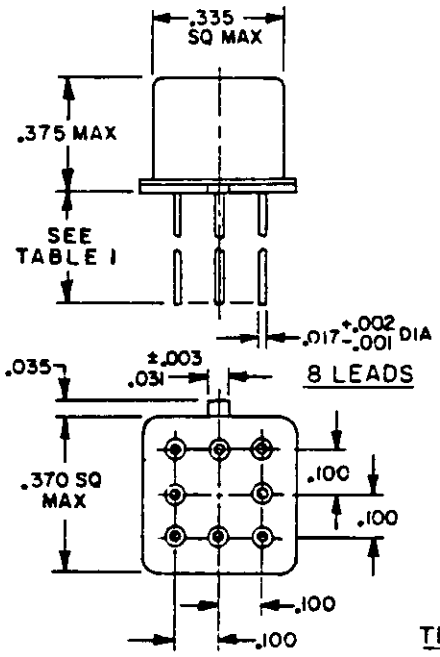
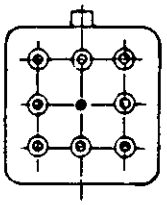
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION TERMINALS 0.100-INCH GRID PATTERN (SENSITIVE 60 MILLIWATTS COIL OPERATE POWER AT 25°C)

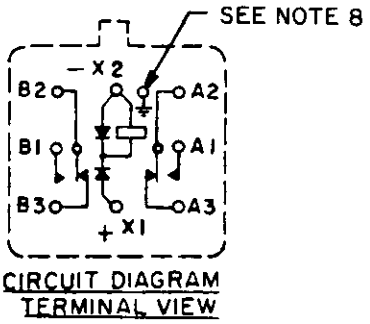
(D) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-39016.





TERMINAL VIEW
-057 THRU -064
(SEE NOTE 8)



CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.335	8.51
.375	9.53

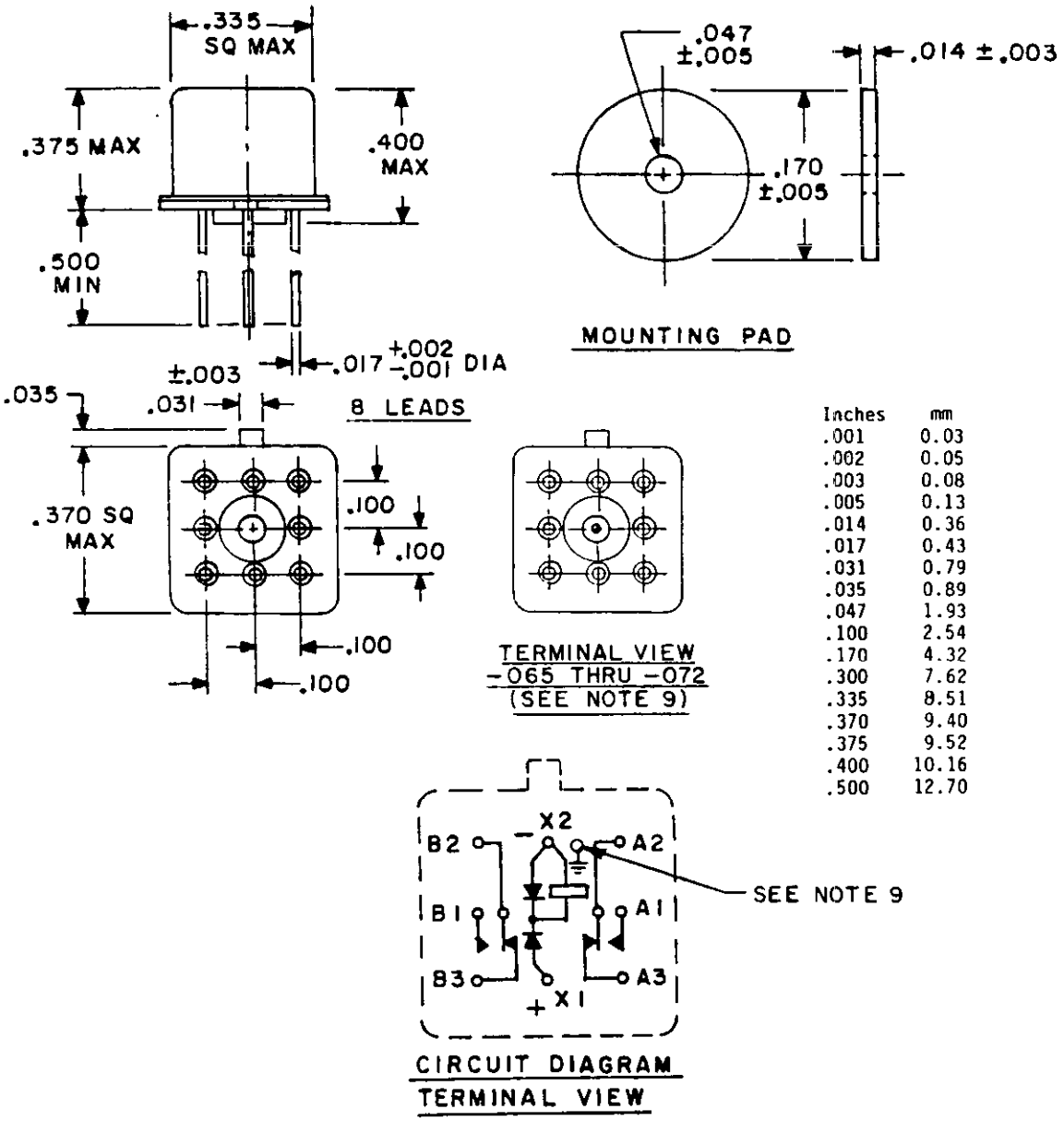
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. The grounding pin shown is a non insulated case grounding pin applicable to -057 through -064 only.

(D) FIGURE 1. Dimensions and configuration.

(D) denotes changes

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.014	0.36
.017	0.43
.031	0.79
.035	0.89
.047	1.93
.100	2.54
.170	4.32
.300	7.62
.335	8.51
.370	9.40
.375	9.52
.400	10.16
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. Mounting pad shall be a polyester film per MIL-I-631, Type G, Class I.
9. The grounding pin shown is a non insulated case grounding pin applicable to -065 through -072 only.

Ⓓ FIGURE 2. Dimensions and configuration (relay with mounting pad).

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(D) Initial: 0.10 ohm maximum (0.110 maximum with spacer pad).

High level:

During life: Not more than 5 percent of circuit voltage.

(D) After life: 0.20 ohm maximum (0.210 maximum with spacer pad).

Low level:

During life: 100 ohms maximum.

(D) After life: 0.15 ohm maximum (0.160 maximum with spacer pad).

Intermediate current:

During: 3 ohms maximum.

(D) After: 0.20 ohm maximum (0.210 maximum with spacer pad).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

(D) Contact stabilization time: 2.5 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

COIL DATA: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 7.5 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance 1/: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	
Between coils of dual coil relays	---	

(D) DIODE CHARACTERISTICS:

(D) Coil transient suppression: Applicable.

(D) Diode breakdown and block integrity (delete coil resistance and substitute the test in all inspection tables of MIL-R-39016):

(D) With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter or oscilloscope or qualifying activity approved test equipmemt. Leakage current shall not exceed the specified value.

Maximum negative transient: 1.0 volt.

(D) Breakdown voltage: 100 V dc minimum at 10 microamperes (μA).

Maximum leakage current: 1 μA at 50 V dc.

(D) Peak inverse voltage: 100 volts minimum.

(D) Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

(D) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

(D) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

(D) Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes.

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PHYSICAL DATA:

Terminals:

Terminal strength: 1 pound pull minimum.

(D) Terminal twist test: As specified in MIL-R-39016.

(D) Dimensions and configuration: See figures 1 and 2.

Terminations: See figure 1 and table I.

Weight: 4.3 grams (0.15 ounce) maximum.

(D) Solderability: Applicable.

(D) Minimum marking: Military part number, "J" with the date code (example J8530), circuit diagram, and manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

(D) Low level: 100,000 cycles plus 900,000 cycles mechanical life.

PART NUMBER: M39016/43- (dash number from table I and suffix letter designating failure rate level).

(D) TABLE I. Dash numbers and characteristics. 1/

Dash number 2/			Coil voltages (V dc) 4/		At +25°C					Over temperature range		
Lead length .500 Min	Lead length .187 ±.010	Mounting pads (fig. 2) 3/	Rated	Max	Coil resistance (ref. only) Ohms 5/	Coil circuit current 5/ (mA) Max	Coil circuit current 5/ (mA) Min	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
033	041	049	5.0	7.0	64	78.1	56.8	2.9	2.2	0.8	3.7	2.6	0.7
034	042	050	6.0	10.0	125	48.9	36.3	4.0	2.5	0.9	4.8	3.0	0.8
035	043	051	9.0	15.0	400	23.6	18.1	6.1	3.6	1.1	8.0	4.5	0.9
036	044	052	12.0	20.0	800	16.0	12.5	7.8	4.6	1.3	11.0	5.8	1.0
037	045	053	18.0	30.0	1,600	12.2	9.6	11.3	7.0	1.5	14.5	9.0	1.1
038	046	054	26.5	40.0	3,200	9.0	7.2	15.2	10.8	1.7	19.0	13.0	1.3
039	047	055	36.0	57.0	6,500	6.1	4.9	21.7	14.7	2.3	27.2	19.0	1.7
040	048	056	48.0	75.0	11,000	4.8	3.9	27.8	19.8	2.8	34.8	26.0	2.0
057 6/	---	065 6/	5.0	7.0	64	78.1	56.8	2.9	2.2	0.8	3.7	2.6	0.7
058 6/	---	066 6/	6.0	10.0	125	48.9	36.3	4.0	2.5	0.9	4.8	3.0	0.8
059 6/	---	067 6/	9.0	15.0	400	23.6	18.1	6.1	3.6	1.1	8.0	4.5	0.9
060 6/	---	068 6/	12.0	20.0	800	16.0	12.5	7.8	4.6	1.3	11.0	5.8	1.0
061 6/	---	069 6/	18.0	30.0	1,600	12.2	9.6	11.3	7.0	1.5	14.5	9.0	1.1
062 6/	---	070 6/	26.5	40.0	3,200	9.0	7.2	15.2	10.8	1.7	19.0	13.0	1.3
063 6/	---	071 6/	36.0	57.0	6,500	6.1	4.9	21.7	14.7	2.3	27.2	19.0	1.7
064 6/	---	072 6/	48.0	75.0	11,000	4.8	3.9	27.8	19.8	2.8	34.8	26.0	2.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 033L - - - - - 056R.

3/ Relays supplied with mounting pads (-049 through -056 and -065 through -072) shall have the pad rigidly attached.

4/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to the coil terminals, the coil circuit current must be within the limits shown. Measure at 25°C at nominal voltage for 5 seconds, maximum.

6/ Relays are supplied with a case grounding pin welded to the header (see figures 1 and 2).

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QUALIFICATION INSPECTION AND SAMPLE SIZE: See table II.

(D) TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable	M39016/43-038	18 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable
	M39016/43-033 M39016/43-034 M39016/43-035 M39016/43-036 M39016/43-037 M39016/43-039 M39016/43-040	2 units each part number qualification inspection, group II
	M39016/43-062	1 unit terminal strength and solderability

1/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

(D) Initial qualification of relays supplied with mounting pads (-049 through -056 and -065 through -072), shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

(D) Before installation of pad; screening, visual and mechanical inspection (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

(D) Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

(D) Qualification inspection (reduced testing for previously qualified relays) for relays supplied with mounting pads (-049 through -056 and -065 through -072), two units of the 26.5 volt rated coil voltage (-052) shall be tested as specified below:

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

For failure rate level L only. Screening.

(D) For failure rate levels M, P, and R: Vibration (sinusoidal), vibration (random), test duration shall be 10 minutes, particle impact noise detection (P.I.N.D., when specified), screening.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

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(D) Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

(D) Group A testing for relays supplied with spreader pads (-049 through -056 and -065 through -072) shall be tested as specified below:

(D) Before installation of pad perform subgroup 2 of group A tests.

(D) After installation of pad perform subgroups 3 and 4 of group A tests.

SUPERSESSION DATA:

Supersession data: See table III.

(D) TABLE III. Supersession data. 1/

Superseded part no. M39016/43-	New part no. M39016/43-	Superseded part no. M39016/43-	New part no. M39016/43-
001	033	017	033
002	034	018	034
003	035	019	035
004	036	020	036
005	037	021	037
006	038	022	038
007	039	023	036
008	040	024	040
009	041	025	041
010	042	026	042
011	043	027	043
012	044	028	044
013	045	029	045
014	046	030	046
015	047	031	047
016	048	032	048

1/ Dash numbers -001 through -032, .350 inch high cans have been canceled and superseded by -033 through -048, .375 inch high cans. The .350 inch high cans are no longer manufactured.

For Government logistical support: See table IV.

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(D) TABLE IV. Cross reference for Government logistical support.

Superseded part no. M39016/43-	New part no. M39016/43-	Support with part no. M39016/43-	Superseded part no. M39016/43-	New part no. M39016/43-	Support with part no. M39016/43-	New part no. M39016/43-	Support with part no. M39016/43-
001	033	033	029	045	037	057	057
002	034	034	030	046	038	058	058
003	035	035	031	047	039	059	059
004	036	036	032	048	040	060	060
005	037	037		033	033	061	061
006	038	038		034	034	062	062
007	039	039		035	035	063	063
008	040	040		036	036	064	064
009	041	033		037	037	065	065
010	042	034		038	038	066	066
011	043	035		039	039	067	067
012	044	036		040	040	068	068
013	045	037		041	033	069	069
014	046	038		042	034	070	070
015	047	039		043	035	071	071
016	048	040		044	036	072	072
017	033	033		045	037		
018	034	034		046	038		
019	035	035		047	039		
020	036	036		048	040		
021	037	037		049	049		
022	038	038		050	050		
023	039	039		051	051		
024	040	040		052	052		
025	041	033		053	053		
026	042	034		054	054		
027	043	035		055	055		
028	044	036		056	056		

CONCLUDING MATERIAL

(D) Custodians:
Army - ER
Navy - EC
Air Force - 85

(D) Review activities:
Navy - AS
Air Force - 99
DLA - ES
NSA - NS

(D) User activities:
Army - AR
Navy - MC, OS, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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